

DATA STRUCTURES AND ALGORITHMS (DSA) COURSE CONTENT

Duration- 6-8 Weeks

WEEK - 01

Introduction & Time Complexity

- What is DSA? Why is it important?
- Asymptotic Analysis: Big O, Big Ω , Big Θ
- Time and Space Complexity
- Recursion and Backtracking Basics
- Practice: Fibonacci, Factorial, Tower of Hanoi

WEEK - 02

Arrays and Strings

- Arrays: Insertion, Deletion, Searching, Traversing
- Prefix Sum, Sliding Window Technique
- 2D Arrays & Matrix Operations
- Strings: Pattern Matching, Palindromes, Anagram checks
- Practice: Kadane's Algorithm, Subarrays with Sum K

WEEK - 02

Linked Lists

- Singly, Doubly, and Circular Linked Lists
- Operations: Insertion, Deletion, Reversal
- Detect Loop using Floyd's Algorithm
- Merge Two Sorted Linked Lists
- Practice: LRU Cache (Design question)

WEEK - 04

Stacks and Queues

- Stack using arrays and linked list
- Infix to Postfix conversion, Valid Parentheses
- Queue and Circular Queue
- Deque and Priority Queue
- Practice: Next Greater Element, Sliding Window Maximum

WEEK - 05

Trees and Binary Search Trees

- Binary Tree: Traversals (Inorder, Preorder, Postorder)
- BST operations: Insertion, Deletion, Search
- Height, Diameter, Lowest Common Ancestor (LCA)
- Binary Heap (Min/Max) and HeapSort
- Practice: Convert BST to Balanced BST

WEEK - 06

Hashing and Searching/Sorting Algorithms

- Hash Tables and HashMaps (with collision resolution)
- Searching: Binary Search and variations
- Sorting Algorithms: Bubble, Merge, Quick, Count, Radix
- Practice: Top K frequent elements, Pair Sum problem

OPTIONAL ADVANCED WEEKS

WEEK - 07

Graphs and Greedy Algorithms

- Graph Representations (Adjacency List/Matrix)
- BFS, DFS, Cycle Detection
- Dijkstra's Algorithm, Prim's and Kruskal's (MST)
- Greedy Algorithms: Activity Selection, Huffman Coding
- Practice: Graph traversal challenges on LeetCode

WEEK - 08

Dynamic Programming & Final Project

- DP Concepts: Memoization vs Tabulation
- Classic Problems: Knapsack, Longest Subsequence, Matrix Chain
- Bit Manipulation Basics
- Practice: DP on Strings, Grids, Subsets
- Capstone Project: Real-world problem using DSA
- Final Assessment & Feedback

Languages & Tools

- **Languages:** C++, Java, or Python
- **Platforms:** LeetCode, HackerRank, Codeforces, GeeksforGeeks
- **IDE Tools:** VS Code, IntelliJ, Jupyter Notebook

Course Deliverables

- Lecture Slides & Code Examples
- Weekly Quizzes & Assignments
- Final Capstone Project
- Interview Preparation Sheet
- Certificate of Completion (if applicable)